

15.0 Milestone Assessment: Routing, Network Segmentation, and Network Tracing

Routing is the process of forwarding IP traffic between networks. Routers operate primarily at Layer 3 of the OSI model and use destination IP addresses and routing tables to determine where packets should be forwarded. When a host communicates outside its local subnet, it sends the packet to its default gateway. The router examines the destination address, selects the best matching route, and forwards the packet through the appropriate interface or next-hop router.

Characteristics of Routing Technologies

Routing information can come from directly connected networks, static routes, default routes, or dynamic routing protocols.

Static routing is manually configured by an administrator. It provides predictable paths, requires little protocol overhead, and is appropriate for small, stable networks, single-path branches, or specific backup routes. Its primary limitation is scalability because routes must be manually changed when the topology changes.

A **default route** provides a path when no more-specific route exists. In IPv4, the default route is `0.0.0.0/0` and is commonly used by a branch or edge router to forward unknown destinations toward an upstream router or ISP.

Dynamic routing allows routers to exchange routing information and automatically adapt when network topology changes. This makes dynamic routing more scalable for larger or redundant networks, although it introduces additional configuration and processing overhead.

Static vs. Dynamic Routing

Characteristic	Static Routing	Dynamic Routing
Route creation	Administrator configures routes	Routers learn routes
Changes	Manual	Automatically adapts
Overhead	Very low	Higher
Scalability	Limited	High

Characteristic	Static Routing	Dynamic Routing
Typical use	Small/stable or specific paths	Large/changing networks
Examples	Static/default route	RIP, OSPF, EIGRP, BGP

Routing Protocols

Routing protocols determine how routers learn and select paths.

Protocol	Type	Primary Metric/Decision	Network+ Concept
RIP	Distance-vector IGP	Hop count	Simple; maximum 15 usable hops
OSPF	Link-state IGP	Cost	Builds a topology database and uses SPF
EIGRP	Advanced distance-vector IGP	Composite metric, primarily bandwidth and delay	DUAL and fast convergence
BGP	Path-vector/interdomain	Path attributes and policy	Routing between autonomous systems

RIP is simple but has limited scalability because it uses hop count as its primary metric. OSPF provides a more scalable approach by exchanging link-state information, building a link-state database, and using the Shortest Path First algorithm to calculate routes. EIGRP uses the Diffusing Update Algorithm (DUAL) and a composite metric, providing more sophisticated path selection than RIP. BGP is different because it is used primarily for interdomain routing between autonomous systems and relies on path attributes and routing policy.

Routing Tables and Forwarding Decisions

A routing table contains the routes a router knows and uses to make forwarding decisions. Entries may include directly connected networks, static routes, dynamically learned routes, and a default route. When determining the best route, the router first uses the longest prefix match, selecting the most specific destination network.

If multiple routes exist for the same prefix from different routing sources, administrative distance is used to determine the more trusted source. The routing protocol's metric is then used to select the preferred path among routes learned by that protocol.

1. **Prefix length** = which is the most specific destination

For example: If a router has routes for `10.0.0.0/8`, `10.10.0.0/16`, and `10.10.20.0/24`, a packet destined for `10.10.20.25` matches all three networks, but the `/24` route is the most specific and is therefore selected.

2. **Administrative distance** = which routing source is more trusted

For example: Suppose a router learns `10.20.0.0/16` through both OSPF and RIP. OSPF has an administrative distance of 110, while RIP has an administrative distance of 120. Because a lower administrative distance is preferred, the router selects the OSPF route.

3. **Metric** = which path the protocol prefers

For example: Suppose a router learns the same `10.30.0.0/16` destination through two OSPF paths. One path has an OSPF cost of 10 and the other has a cost of 50. OSPF selects the path with the lower cost of 10.

Exam memory aid:

- Prefix length → Which destination is most specific?
- Administrative distance → Which routing source is more trusted?
- Metric → Which path does that routing protocol prefer?

Characteristics of Network Segmentation

Network segmentation divides a larger network into smaller physical or logical network segments. Common segmentation technologies include Virtual Local Area Networks (VLANs), IP subnets, and Layer 3 routing. A VLAN creates a separate Layer 2 broadcast domain.

At Layer 3, subnetting divides an IP network into smaller networks, with each subnet having its own network address and address range. Traffic between different VLANs or subnets must be routed. For example, an organization could separate its network into:

- Sales VLAN/subnet
- Human Resources VLAN/subnet
- IT VLAN/subnet
- Server VLAN/subnet
- Guest VLAN/subnet
- Management VLAN/subnet

This segmentation provides several benefits:

1. **Performance:** Smaller broadcast domains reduce unnecessary broadcast traffic.

2. **Security:** Sensitive systems can be isolated from general users. ACLs and firewalls can control which segments are permitted to communicate.
3. **Management:** Departments and device types can be logically organized and managed independently.
4. **Troubleshooting:** A problem can be isolated to a particular VLAN, subnet, or network segment instead of affecting the entire network.
5. **Routing:** Communication between segments requires Layer 3 routing. This can be provided by a router, router-on-a-stick configuration, or a Layer 3 switch using switched virtual interfaces (SVIs).

Segmentation therefore connects directly to routing in that VLANs and subnets create separate networks, while routers or Layer 3 switches provide controlled communication between those networks. The Network+ exam objectives specifically emphasize VLANs, subnetting, and the need to route traffic between segmented networks.

Characteristics of Tracing Network Activities

Tracing network activities involves following network communication to determine where traffic is traveling, whether a destination is reachable, and where a connectivity or routing problem may occur. Several tools can be used:

Tool	Purpose	Example Use
ping	Tests IP connectivity and reachability	Determine whether a host or gateway responds
tracert/traceroute	Shows the Layer 3 path toward a destination	Identify where a route fails or loops
Wireshark	Captures and analyzes packets	Examine protocols, addresses, ports, and conversations
Nmap	Discovers hosts, ports, and services	Identify reachable devices and exposed services
Routing-table commands	Display learned and configured routes	Determine how a router will forward traffic

A systematic troubleshooting process can begin with the ping command. A technician can test the local TCP/IP stack, the default gateway, another local host, and finally a remote destination. If local connectivity works but a remote destination cannot be reached, traceroute/tracert can identify the routers or hop where communication stops or takes an unexpected path. CompTIA's

Network+ study material specifically describes using traceroute to investigate routing problems and repeated addresses that may indicate a routing loop.

Wireshark provides deeper packet-level visibility. A technician can inspect source and destination IP addresses, protocols, ports, TCP handshakes, retransmissions, Domain Name Service (DNS) requests, and other packet behavior. This helps determine whether a problem is related to routing, DNS, TCP communication, or another protocol.

Nmap can be used to identify reachable hosts and discover open ports and services. This is useful for network discovery, troubleshooting, and identifying unexpected services that may require further security investigation.

Routing-table information is also important when tracing traffic. Examining a router's routing table can reveal whether the destination network is known, which next hop will be used, and whether a default route is being relied upon.

Practical Troubleshooting Example

Suppose a workstation can communicate with devices on its local VLAN but cannot reach a server on another subnet. This process allows a technician to move from basic connectivity testing to path analysis and finally to packet-level investigation.

1. **Ping the workstation's own address** to verify the local TCP/IP stack.
2. **Ping the default gateway** to verify local Layer 3 connectivity.
3. **Ping another host on the same VLAN** to verify local network communication.
4. **Ping the remote server** to determine whether inter-network communication is working.
5. **Run tracert/traceroute** to identify where the path fails.
6. **Examine routing tables** on the relevant router or Layer 3 switch to determine whether the destination network has a valid route.
7. **Use Wireshark** as needed, to determine what traffic is actually occurring.
8. **Use Nmap**, if appropriate and authorized, to investigate reachable hosts, ports, & services.

Key Takeaway

Routing determines where traffic goes. Segmentation determines which networks are separated. Tracing tools determine what is happening along the path. Together, these concepts allow a network technician to design segmented networks, select appropriate routing technologies, understand router forwarding decisions, and systematically troubleshoot connectivity problems.